Rafael Arraez

Senior Backend Engineer | AI Agents & LLMs | Node.js · NestJS · TypeScript · PostgreSQL
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PROFESSIONAL SUMMARY

Senior Backend Engineer with 8+ years designing, building and scaling backend platforms with Node.js, NestJS, TypeScript and Python/Django. Specialized in LLM-powered AI agents (LangGraph) that quote and book for 700+ hotels, and in taking backends from zero to production: architecture (Clean Architecture, DDD, microservices and event-driven), real-time (WebSockets and WebRTC), security, CI/CD and observability on AWS and Google Cloud.

PROFESSIONAL EXPERIENCE

Software Engineer | WeSpeak Apr 2026 – Present | USA (Remote)
AI-powered WhatsApp sales platform for 700+ hotels in 20 countries.

- Built 10+ integrations with PMSs and booking engines (Lobby PMS, Silbeck, Niara, Anfy, OYO, GuruHotel, Paraty Tech and Tuzgle) so the assistant can quote and book in real time.
- Built the conversion attribution system (pixel) end to end: embeddable JavaScript tag, UTM tracking, events in TinyBird and an ROI dashboard, with automatic promotion of leads to the won stage that removed manual work for the sales team.
- Developed the Web Chat channel: an embeddable widget connected to the AI assistant, with payment links and prices in the visitor's local currency.
- Implemented the TravelCash cashback flow, triggered by payment gateway webhooks, with three layers of idempotency and durable jobs in Inngest.
- Improved the accuracy of the LangGraph conversational agent: quotes, occupancy calculation and same-turn handoff to human agents.

Tech stack: TypeScript, Node.js, Next.js, LangGraph, WhatsApp Cloud API, MySQL, Redis, Inngest, Vercel.

Lead Backend Engineer | SilverConnect Oct 2025 – Present | Valencia, Spain (Remote, Contract)
AgeTech startup: mobile app (iOS and Android) providing remote tech support for older adults.

- Designed and built the NestJS backend from scratch with Clean Architecture and DDD, plus the Next.js admin dashboard, through the MVP launch on the App Store and Google Play.
- Built real-time remote assistance: WebRTC signaling over WebSockets (Socket.IO) for screen sharing and remote control, with session recovery after disconnections.
- Implemented REST APIs with JWT authentication and rotating refresh tokens, 2FA (TOTP), Google sign-in, role-based access control and push notifications with Firebase.
- Designed the PostgreSQL data model with TypeORM, indexing frequent queries, and removed N+1 queries in metrics endpoints (from 2 queries per user to a single grouped query).
- Deployed to Google Cloud Run with Docker and CI/CD on GitHub Actions (tests, lint and rollback), with observability in New Relic and Cloud Logging, alerts and dashboards.

Tech stack: NestJS, TypeScript, PostgreSQL, MongoDB, Socket.IO, WebRTC, Docker, Google Cloud, New Relic.

Senior Software Engineer | Magellan Feb 2024 – Oct 2025 | USA (Remote)

- Designed and implemented large-scale integrations between HubSpot, Salesforce and SalesLoft, syncing millions of marketing and sales records.
- Designed NestJS and Django APIs for an email personalization SaaS platform with 200K+ monthly active users.
- Built backend modules with Django and the OpenAI API to automate email marketing campaigns, increasing conversion by 25%.
- Optimized complex PostgreSQL queries, cutting response times by 60%, and built CI/CD pipelines on AWS with Docker that reduced deployments from 2 h to 15 min.

Senior Software Engineer | WiTi Nov 2022 – Feb 2024 | Chile (Remote)

- Implemented a multi-store order management system in NestJS and AWS (Lambda, API Gateway, DynamoDB) as part of a monolith-to-microservices migration, processing 1M+ transactions per month.
- Cut infrastructure costs by 20% by optimizing AWS services.
- Raised test coverage to 85% with Jest across backend (NestJS) and frontend (Next.js).
- Mentored 3 junior developers on code reviews and NestJS and AWS best practices, shortening their ramp-up by 40%.

Full Stack Developer | Raxo Creative Feb 2021 – Nov 2022 | USA (Remote)

- Led development of a social network in Next.js and NestJS with real-time notifications, admin roles, JWT authentication and dynamic permission middleware.
- Implemented SSR and SSG in Next.js, improving SEO and load time by 40%.

Full Stack Developer | AKB Fintech Jan 2020 – Feb 2021 | USA (Remote)

- Integrated Stripe payments for 50K+ active users and built bulk SMS and email delivery with Twilio and Mailgun, improving deliverability by 35%.
- Built an omnichannel support system in Laravel, Vue.js and PostgreSQL and refactored its queries, reducing latency by 30%.

Full Stack Developer | Impulsa Creativos Jul 2018 – Dec 2019 | Valencia, Venezuela

- Built REST APIs in Laravel for e-commerce and landing pages in Vue.js.

EDUCATION

Bachelor's Degree in Computer Engineering | Universidad José Antonio Páez, Venezuela | Jul 2019

TECHNICAL SKILLS

Programming languages: TypeScript, JavaScript, Python, PHP, SQL
Backend: Node.js, NestJS, Django, Laravel, REST APIs, WebSockets (Socket.IO), WebRTC
AI & LLMs: LangGraph, LangChain, OpenAI API, AI agents, WhatsApp Cloud API
Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, TypeORM
Cloud & DevOps: AWS (Lambda, API Gateway, S3, RDS, SQS, SNS), Google Cloud Run, Docker, GitHub Actions, Vercel
Architecture & quality: Clean Architecture, DDD, microservices, event-driven, serverless, TDD, Jest, Vitest
Frontend: Next.js, React, Vue.js

LANGUAGES

Spanish: Native | English: Intermediate (B1)